

№6

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SIR ()

$$\begin{aligned}
 (t = 0) \quad & \left(\begin{array}{l} \text{.} \quad , \quad (N = 10900) \\ I(0) = 210, \\ R(0) = 43. \quad , \\ S(0) = N - I(0) - R(0). \end{array} \right.
 \end{aligned}$$

$$1) \quad I(0) \leq I^*;$$

$$2) \quad I(0) > I^*.$$

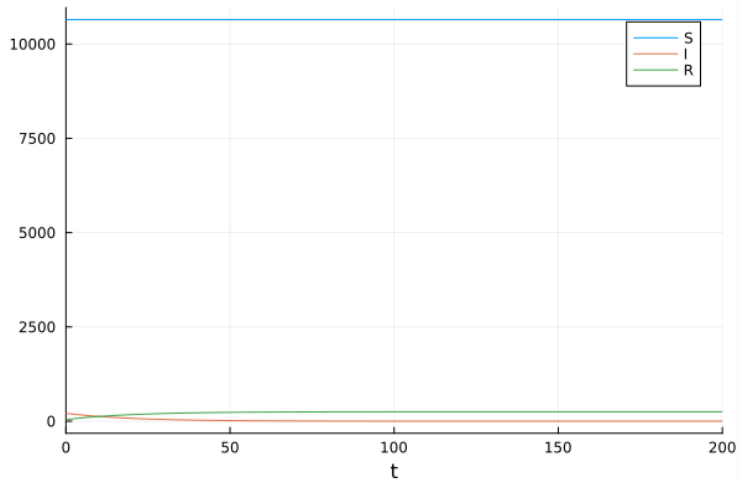
$$I(0) \leq I^*$$

```
function sir_2(u,p,t)
    (S,I,R) = u
    (b, c) = p
    N = S+I+R
    dS = 0
    dI = -c*I
    dR = c*I
    return [dS, dI, dR]
end
```

```
N = 10900
I_0 = 210
R_0 = 43
S_0 = N - I_0 - R_0
u0 = [S_0, I_0, R_0]
p = [0.1, 0.05]
tspan = (0.0, 200.0)
```



```
prob_2 = ODEProblem(sir_2, u0, tspan, p)
sol_2 = solve(prob_2, Tsit5(), saveat = 0.1)
plot(sol, label = ["S" "I" "R"])
```

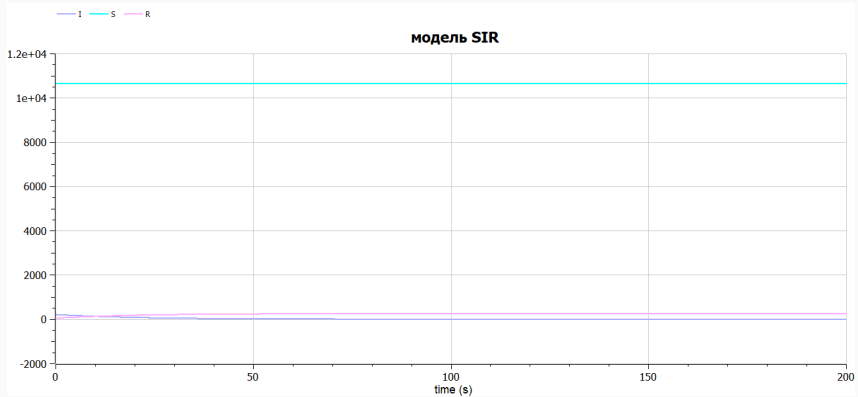


. 1:

```
parameter Real I_0 = 210;  
parameter Real R_0 = 43;  
parameter Real S_0 = 10647;  
parameter Real N = 10900;  
parameter Real b = 0.1;  
parameter Real c = 0.05;  
Real S(start=S_0);  
Real I(start=I_0);  
Real R(start=R_0);
```

equation

```
der(S) = 0;  
der(I) = - c*I;  
der(R) = c*I;
```



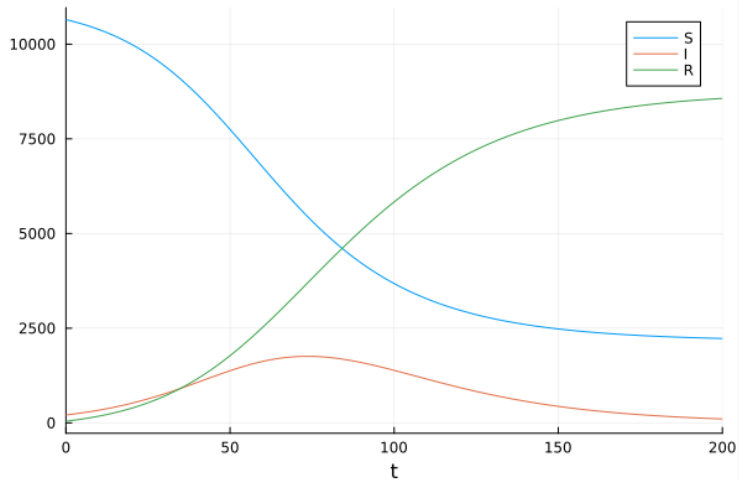
. 2:

$$I(0) > I^*$$

```
function sir(u,p,t)
    (S,I,R) = u
    (b, c) = p
    N = S+I+R
    dS = -(b*S*I)/N
    dI = (b*I*S)/N - c*I
    dR = c*I
    return [dS, dI, dR]
end
```

```
N = 10900
I_0 = 210
R_0 = 43
S_0 = N - I_0 - R_0
u0 = [S_0, I_0, R_0]
p = [0.1, 0.05]
tspan = (0.0, 200.0)
```

```
prob = ODEProblem(sir, u0, tspan, p)
sol = solve(prob, Tsit5(), saveat = 0.1)
plot(sol, label = ["S" "I" "R"])
```

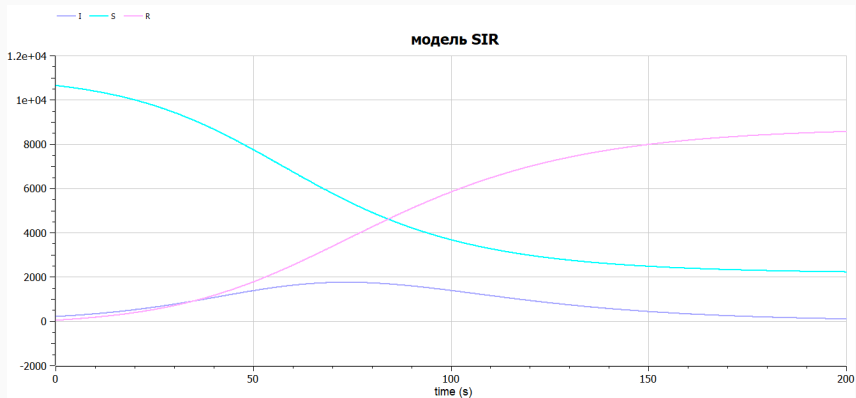



. 3:

```
parameter Real I_0 = 210;  
parameter Real R_0 = 43;  
parameter Real S_0 = 10647;  
parameter Real N = 10900;  
parameter Real b = 0.1;  
parameter Real c = 0.05;  
Real S(start=S_0);  
Real I(start=I_0);  
Real R(start=R_0);
```

equation

```
der(S) = -(b*S*I)/N;  
der(I) = (b*I*S)/N - c*I;  
der(R) = c*I;
```



. 4:

SIR.

1. Compartmental models in epidemiology []. URL: https://en.wikipedia.org/wiki/Compartmental_models_in_epidemiology.